

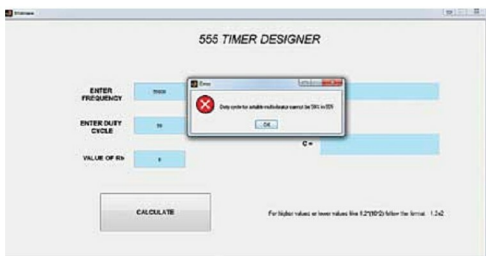
Fig. 6: Flow-chart for the MATLAB program

Thus either the external voltage or a potmeter connected to this pin determines the pulse width of the output waveform. The external voltage applied to this pin can also be used to modulate the output waveform. When this pin is not used, it should be connected to ground through a $0.01\mu\text{F}$ capacitor to avoid any noise problem.

Pin 6: Threshold terminal. This is the non-inverting input terminal of comparator-1. It compares the voltage applied to the terminal with a reference voltage of $2/3V_{cc}$. The amplitude of the voltage applied to this terminal is responsible for set state of the output



Fig. 7: Values entered in the designer

Fig. 8: Error for duty cycles ≤ 50

of flip-flop.

Pin 7: Discharge terminal. This pin is connected internally to the collector of the transistor; mostly a capacitor is connected between this terminal and the ground. It is called discharge terminal because when the transistor saturates, the capacitor discharges through the transistor. The capacitor charge/discharge rate is determined by the external resistors.

Pin 8: Supply terminal. A supply voltage of $+5\text{V}$ to $+18\text{V}$ is applied to this terminal with respect to the ground (pin 1).

There are literally thousands of different ways in which the 555 timer can be used in electronic circuits. In almost every case, however, the basic circuit is wired in one of the three operating modes:

Monostable mode. In this mode, the 555 timer acts as 'one-shot pulse generator,' producing a single pulse when triggered. As shown in Fig. 3, the external capacitor is initially held discharged by a transistor inside the timer. The circuit triggers on a negative-going input signal when its level reaches $1/3V_{cc}$. Once triggered, the



For More
Channel Related Stories
 Visit www.electronicshobby.com